

Ruben Carrazco

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Education

Stanford University

Stanford, CA

B.S. Electrical Engineering (Hardware and Software Track)

Sep. 2023 - Present

Cumulative GPA: 3.79/4.00

Courses: Circuit Design, Digital System Design, Intro to VLSI Systems, Probability and ML Foundations

Experience

CS 198 - Student Leader (CS106A/B)

Sept. 2025 - Present

Stanford University

- Teach and mentor students in Python (CS106A) and C++ (CS106B) through weekly sections, emphasizing algorithmic thinking, abstraction, and debugging best practices.

Stanford Student Space Initiative (SSI) - Avionics Team Member

Apr. - June 2025

Stanford University

- Developed an I2C-based gas control system using a Teensy microcontroller to monitor real-time pressure and temperature inside the rocket's tank system.
- Programmed Arduino IDE drivers for solenoid valves and co-designed a modular electronics stand to verify bidirectional communication across all subsystems.

Projects

Wearable VR Glove

May 2025 - Present

Stanford University | EE267, Virtual Reality

- Built a wireless microcontroller-based glove integrating flex sensors and an IMU for real-time gesture-based control in Unity, achieving <50 ms latency and seamless head-tracked interaction.
- Programmed a Teensy-ESP32 system using ESP-NOW and quaternion-based tracking, reaching ~86% average gesture classification accuracy without reliance on cameras or external controllers.

AC to DC Converter

Jan. - Feb. 2025

Stanford University | EE101A, Circuits I

- Implemented a full-bridge AC-DC converter with capacitive filtering to reduce ripple and regulate output voltage under various load conditions.
- Measured performance using oscilloscopes and waveform generators; validated and refined transient behavior through iterative LTspice simulations.

Enhanced Music Synthesizer & Display

Jan. - Feb. 2025

Stanford University | EE108, Digital Design

- Created a Verilog-based music synthesizer for FPGA capable of waveform mixing, harmonic generation, and amplitude control displayed via VGA output.
- Debugged timing and control logic using Xilinx Vivado for glitch-free output on VGA hardware.

SIMD GEMM Accelerator (In Progress)

Sep. 2025 -

Present

Stanford University | EE271, Introduction to VLSI Systems

- Writing C/C++ and SystemVerilog code to implement and synthesize a SIMD GEMM accelerator using EDA tools, with performance optimization through high-level synthesis.

Skills

Hardware & Electronics: Circuit Design, Signal Processing, Sensor Integration, Soldering

Programming: Embedded C/C++, Verilog, Python, Arduino IDE, Assembly (x86-64), I2C, Unity C#

Tools & Software: LTspice, Xilinx Vivado, KiCad, Fusion 360, Logic Analyzers, Oscilloscopes, Multimeters

Languages: English, Spanish (Working Proficiency)